

WASP-12b

R-2244

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150104 · created 2026-08-15T23:21:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457026.8834 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.048 +/- 0.03
Transit depth (Rp/Rs)^2	0.23 +/- 0.29 [%]
Orbital Inclination (inc)	86.29 +/- 2.88
Airmass coefficient 1 (a1)	0.967 +/- 0.012
Airmass coefficient 2 (a2)	0.0198 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#2 - [269, 312]
Optimal Method	PSF photometry
Transit Duration (day)	0.1028 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.048 $\pm$ 0.03 vs 0.1178 (deviation 2.32 σ)
Duration fit vs published	0.1028 d vs 0.125 d (deviation 8.87 σ)
Mid-time O–C	0.3 min
Depth SNR	1.6
Residual scatter	1.24 %

Light curve

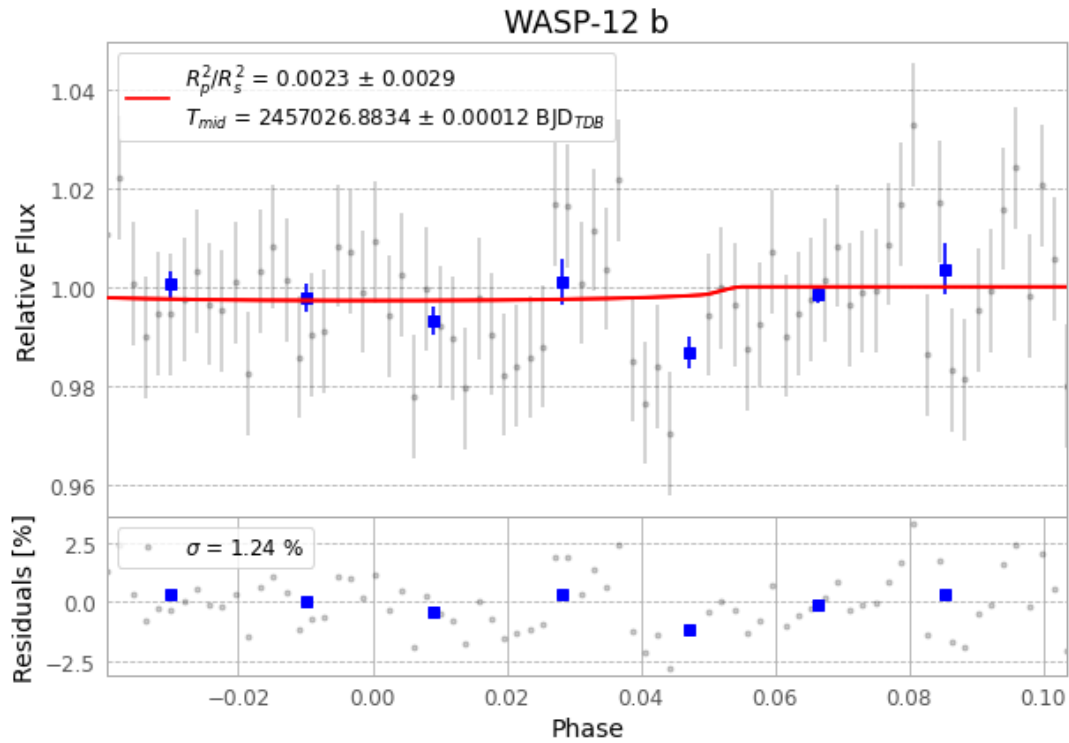


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-04.png (SHA-256 cfa6068f68fdf25e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fb6af27738f41b96...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

74 FITS frames (49 MB), WASP-12150104080013.FITS → WASP-12150104114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150104.log (2b1405f5ee80...), FinalLightCurve_WASP-12 b_2015-01-04.png (cfa6068f68fd...), FinalLightCurve_WASP-12 b_2015-01-04.csv (fcae00cca879...)

Compute resources

Wall time 112.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:21Z.